RAJALAKSHMI INSTITUTE OF TECHNOLOGY

ME23312 ENGINEERING THERMODYNAMICS

UNIT I – Basics, Zeroth and First Law • Lecture Notes

SESSION 9 • LECTURE 7

First Law for Open Systems — The Steady Flow Energy Equation

Detailed lecture notes — control volumes, flow work, enthalpy, SFEE derivation, solved numericals

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Learning objectives. After this session you should be able to: (i) define control volume, control surface and steady flow; (ii) explain flow work and enthalpy; (iii) derive the SFEE and state its assumptions; (iv) use both the rate form and the per-unit-mass form with consistent units; (v) compute enthalpy and flow work from p, v, u data.

1. Control Volume Analysis

Devices such as turbines, compressors, boilers and nozzles operate with continuous mass flow: the closed-system (control-mass) method fails. Instead we fix a region in space — the **control volume** (CV) — bounded by a **control surface** (CS) across which mass, heat and work may all pass.

- **Steady flow:** no property at any point inside the CV changes with time; the mass and energy contained are constant.
- **Mass balance (continuity):** $\dot{m}_1 = \dot{m}_2 = \dot{m} = \rho AV = AV/v$, where V is velocity and v the specific volume.
- Because stored energy is constant, the rate of energy in must equal the rate of energy out.

2. Flow Work and Enthalpy

To push each kilogram of fluid across the control surface against the local pressure p, the fluid upstream must do work on it. For a plug of volume v entering through area A: force = pA, distance = v/A, so work = pv. This **flow work** (flow energy) accompanies every kilogram crossing the CS.

$$h = u + pv \quad (\text{kJ/kg})$$

Enthalpy bundles internal energy with flow work; a flowing stream carries $h + V^2/2 + gz$ per kg

KEY POINT Use u for energy stored in a non-flow (closed) system; use h for energy transported by flowing streams. This single habit prevents most open-system errors. Note h is a property — defined for stationary fluids too.

3. Derivation of the SFEE

1. Assumptions: steady state, one inlet and one outlet, one-dimensional flow at the ports, no accumulation of mass or energy.
2. Rate of energy in: $\dot{Q} + \dot{m}(h_1 + V_1^2/2 + gz_1)$.
3. Rate of energy out: $\dot{W} + \dot{m}(h_2 + V_2^2/2 + gz_2)$.
4. Steady state ⇒ in = out. Rearranging gives the Steady Flow Energy Equation.

$$\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)]$$

Rate form: $\dot{Q}$, $\dot{W}$ in kW; $\dot{m}$ in kg/s; h in kJ/kg ⇒ divide $V^2/2$ and gz by 1000

$$q - w = (h_2 - h_1) + (V_2^2 - V_1^2)/2000 + g(z_2 - z_1)/1000$$

Per unit mass, all terms in kJ/kg (V in m/s, z in m)

Term	Physical meaning	Unit (per-mass form)
q	Heat to the fluid per kg	kJ/kg
w	Shaft work by the fluid per kg	kJ/kg
$h_2 - h_1$	Enthalpy change (u + pv)	kJ/kg
$(V_2^2 - V_1^2)/2000$	Kinetic-energy change	kJ/kg
$g(z_2 - z_1)/1000$	Potential-energy change	kJ/kg

KEY POINT $V^2/2$ is in J/kg; divide by 1000 before adding to h in kJ/kg. Omitting this factor is the single most common CAT error. For gases, $g\Delta z$ is nearly always negligible.

4. Worked Examples

WORKED EXAMPLE 7.1 — Enthalpy and flow work of steam

Steam in a pipeline at 800 kPa has specific volume 0.24 m^3/kg and specific internal energy 2576 kJ/kg. Find the flow work per kg and the specific enthalpy.

Solution.

1. Flow work = $pv = 800 \times 0.24 = 192$ kJ/kg.
2. $h = u + pv = 2576 + 192 = 2768$ kJ/kg.

ANSWER $pv = 192$ kJ/kg | $h = 2768$ kJ/kg

Note: kPa × m^3/kg = kJ/kg. Steam-table enthalpies (Unit III) are constructed exactly this way.

WORKED EXAMPLE 7.2 — Mass flow from continuity

Air with specific volume 0.85 m^3/kg flows at 40 m/s through a duct of cross-section 0.03 m^2. Find the mass flow rate.

Solution.

1. $\dot{m} = AV/v = (0.03 \times 40)/0.85$.
2. $\dot{m} = 1.2/0.85 = 1.412$ kg/s.

ANSWER $\dot{m} \approx 1.41$ kg/s

WORKED EXAMPLE 7.3 — Full SFEE with all terms

Fluid enters a device at h_1 = 2800 kJ/kg, V_1 = 50 m/s, z_1 = 10 m and leaves at h_2 = 2600 kJ/kg, V_2 = 150 m/s, z_2 = 4 m. Heat loss is 20 kJ/kg; g = 9.81 m/s^2. Find the specific work output.

Solution.

1. $\Delta h = -200$ kJ/kg.
2. $\Delta KE = (150^2 - 50^2)/2000 = (22500 - 2500)/2000 = +10$ kJ/kg.
3. $\Delta PE = 9.81 \times (4 - 10)/1000 = -0.0589$ kJ/kg (negligible, but keep it here).
4. $q - w = \Delta h + \Delta KE + \Delta PE \Rightarrow -20 - w = -200 + 10 - 0.059 = -190.06$.
5. $w = -20 + 190.06 = 170.06$ kJ/kg (work delivered by the fluid).

ANSWER $w \approx +170.1$ kJ/kg

Note: Observe the relative sizes: Δh dominates; ΔPE is a rounding error for gases and vapours.

5. Practice Problems (answers given)

#	Problem	Answer
P1	Water at 5 MPa has v = 0.001 m^3/kg and u = 1148 kJ/kg. Find h.	1153 kJ/kg
P2	Gas enters a horizontal device at 80 m/s and leaves at 30 m/s with no heat or work. Find Δh.	+2.75 kJ/kg
P3	Steam flows at 3 kg/s through a pipe of area 0.02 m^2 with v = 0.3 m^3/kg. Find the velocity.	45 m/s
P4	Reproduce the SFEE derivation from memory, stating all assumptions.	See Section 3

6. Summary

- Open systems are analysed on a control volume; steady flow means constant stored mass and energy.
- Flow work pv rides with every kg crossing the control surface; h = u + pv absorbs it.
- SFEE: $q - w = \Delta h + \Delta KE + \Delta PE$, with the 2000 and 1000 factors in kJ/kg form.
- The derivation (energy-in = energy-out) with stated assumptions is a standard 8-mark question.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 5 (First Law Applied to Flow Processes). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.